DIEGO CORTÉS MORALES

Postdoctoral Researcher

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WORK EXPERIENCE

Postdoctoral Contract

Mediterranean Institute of Advanced Studies, IMEDEA

Funded by Copernicus Climate Change Service (ECMWF/COPERNICUS/2024/C3S2_520_CNR)

Supervisors: Dr. Ananda Pascual
December 2025 – April 2026 Mallorca, Spain

- Evaluation of the contribution of Eddy Kinetic Energy to Total Kinetic Energy derived from different versions of multi-satellite altimetry products.
- Negotiation and management of funding for research group projects.
- Validation of SWOT observations using gliders in the Mediterranean Sea

Postdoctoral Contract: Validation of Altimetry-Based Geostrophic Velocity Datasets and Generation of Lagrangian Products

Mediterranean Institute of Advanced Studies, IMEDEA

Funded by the European Space Agency "4DMED-SEA" Project (ESA Contract No. 4000141547/23/I-DT)

Supervisors: Dr. Ismael Hernández-Carrasco, Dr. Ananda Pascual
January 2024 – December 2025 Mallorca, Spain

- Evaluation of newly created geostrophic velocities derived from satellite altimetry by integrating in-situ observations through Eulerian and Lagrangian methodologies.
- Generation of Finite-Size Lyapunov Exponents (FSLE) and Finite-Time Lagrangian Vorticity (FTLV) products based on the newly created geostrophic velocities.
- Development of Mediterranean atlas of Rotationally Coherent Lagrangian Vortices (RCLV) using DUACS geostrophic velocities.

PhD in Climate, Atmospheric, Ocean, Terrestrial and Planetary Sciences: Large-Scale Vertical Velocities in the Global Open Ocean via Linear Vorticity Balance

Sorbonne Université, LOCEAN

Funded by the French Ministry of High Education and Research

Supervisors: Dr. Alban Lazar, Dr. Juliette Mignot and Dr. Diana Ruiz Pino
September 2020 – January 2024 Paris, France

- Assessment of regions where Linear Vorticity Balance closes the vorticity budget and implementation of the equation to estimate the vertical velocities within an OGCM NEMO simulation.
- Estimation of vertical velocities (OLIV3) through the utilization of observation-based gridded meridional velocities (ARMOR3D).

ABOUT ME

I am a 30-year-old physical oceanographer and former Postdoctoral Researcher at the Mediterranean Institute of Advanced Studies (IMEDEA, Spain), with extensive experience in large-scale ocean circulation and the assessment and reconstruction of oceanic vertical dynamics from observation-based products. My work has involved the development and evaluation of geostrophic vertical velocity datasets, as well as the application of Lagrangian techniques to altimetry-derived velocity fields for the detection of transport fronts, coherent eddies, and mesoscale circulation features in the Mediterranean Sea.

EDUCATION

PhD. in Climate, Atmospheric, Ocean, Terrestrial and Planetary Sciences

Sorbonne Université, LOCEAN

September 2020 – January 2024

Thesis title: Large-Scale Vertical Velocities in the Global Open Ocean via Linear Vorticity Balance

PhD courses:
- Machine Learning Introduction **(12h)**
- Phytoplanctonic Biomass Spatial Teledetection **(18h)**
- Numerical Modelling and Assimilation of Atmospheric Observations **(20h)**

M.Sc. in Meteorology and Geophysics

Universidad Complutense de Madrid, UCM

September 2019 – July 2020

M.Sc thesis: Impact of the assimilation of surface data for the reconstruction of the North Atlantic Ocean

B.Sc. in Physics

Universidad Autónoma de Madrid, UAM

September 2014 – June 2019

- **Exchange Programme (Erasmus+):** Università degli Studi di Firenze, Italy (09/2016 – 07/2017)

- Comparison of vertical dynamics reproduced by OLIV3 with model, reanalysis and observation-based datasets.

Scientific Internship

Instituto Español de Oceanografía, IEO

January 2020 – May 2020 Madrid, Spain

Analysis and management of oceanographic and meteorological data from research ships. Monthly datasets submitted in EMODnet.

Scientific Internship

Instituto Geográfico Nacional, IGN

January 2019 – April 2019 Madrid, Spain

Comparison of earthquake generated tsunami simulations in the Mediterranean and Atlantic basins and assimilation into early warning systems.

PUBLICATIONS

PhD Thesis

- **D. Cortés-Morales**, *Large-Scale Vertical Velocities in the Global Open Ocean via Linear Vorticity Balance*. 2024. DOI: https://theses.hal.science/tel-04610493v1.

Journal Articles

In preparation

- **D. Cortés-Morales** and I. Hernández-Carrasco, "Role of Coherence Rate in the Stability of Lagrangian Vortices in the Mediterranean Sea," *in preparation*,
- **D. Cortés-Morales** and I. Hernández-Carrasco, "Spatial and Temporal Variability of Eddy Lagrangian Coherence in the Mediterranean Sea over Three Decades of Satellite Altimetry," *in preparation*,
- L. Gómez-Navarro, N. Zarokanellos, B. Barcelo-Llull, *et al.*, "Analysis of daily variability of small-scale structures from gliders and SWOT," *in preparation*,

Submitted

- L. Gómez-Navarro, A. Pascual, **D. Cortés-Morales**, M. Ballarotta, M.-I. Puyol, and L. Fortunato, "New Insights on Mesoscale Activity in the Western Mediterranean Sea," *submitted to State of the Planet*, DOI: https://doi.org/10.5194/sp-2025-17.

Published

- **D. Cortés-Morales**, A. Lazar, D. Ruiz-Pino, and J. Mignot, "Global Thermocline Vertical Velocities: a Novel Observation-Based Estimate," *accepted in Earth System Science Data*, DOI: https://doi.org/10.5194/essd-2025-533.
- **D. Cortés-Morales**, B. Barcellò-Llull, L. Gómez-Navarro, and A. Pascual, "Eddy Kinetic Energy Contribution to Total Ocean Kinetic Energy From Multi-Satellite Altimetry," *Geophysical Research Letters*, 2026. DOI: https://doi.org/10.1029/2026GL123075.

B.Sc thesis: Changes in the upper North Atlantic during XX and XXI centuries.

COMPUTER SKILLS

Proficient

Python | Matlab | Fortran

Intermediate

Lyx/Latex | Github | Git | OceanDataView

Basic

OriginLab | SeaDAS | R | C++

LANGUAGES

Spanish ●●●●●●
Native Speaker

English ●●●●●○
First Certificate (2016)

French ●●●●○○

Italian ●●●●○○

Chinese ●○○○○○
HSK 1 (2019)

FOUNDED PROJECTS

LVBW3D-CLIM

Mercator International Lefe

IPs: Alban Lazar and **Diego Cortés Morales**
January 2022 – December 2023

Founded with 10 000 euros

HARMONY Validation Concept Consolidation and Verification Methodologies

European Space Agency

January 2026 – June 2028

RESEARCH STAYS

PhD Research visit (60 days)

University of California-Los Angeles

Founded by Mercator International Lefe LVBW3D-CLIM project

Supervisor: Jonathan Gula
May – June 2023 California, USA

- **D. Cortés-Morales** and A. Lazar, “Vertical Velocity 3D Estimates from the Linear Vorticity Balance in the North Atlantic Ocean,” *Journal of Physical Oceanography*, 2024. DOI: https://doi.org/10.1175/JPO-D-24-0011.1.

Conference Presentations

- **D. Cortés-Morales** and I. Hernández-Carrasco, “Spatial and Temporal Variability of Eddy Lagrangian Coherence in the Mediterranean Sea over Three Decades of Satellite Altimetry,” in *Ocean Science Meeting 2026*, in-situ presentation, 2026.
- **D. Cortés-Morales** and I. Hernández-Carrasco, “Polarity asymmetry in Lagrangian coherent vortices in the Mediterranean Sea,” in *4th Nonlinear Processes in Oceanic and Atmospheric Flows*, in-situ presentation, 2025.
- **D. Cortés-Morales** and I. Hernández-Carrasco, “How beaching affects the computation and statistics of Lagrangian Coherent Structures,” in *4DMED-Sea Workshop “Beaching Particles and Closed Boundary Conditions”*, in-situ presentation, 2024.
- **D. Cortés-Morales** and A. Lazar, “LVBW-3DClim: Reconstructing the vertical velocities in the global thermocline from observations,” in *LEFE/GMMC Scientific days*, online presentation, 2024.
- **D. Cortés-Morales** and A. Lazar, “Estimating the interannual variability of vertical velocities within the global ocean thermocline from observation-based geostrophic meridional velocities,” in *EGU General Assembly*, online presentation, 2023.
- **D. Cortés-Morales** and A. Lazar, “LVBW-3DClim: Estimating the interannual variability of vertical velocities within the global ocean thermocline from observation-based geostrophic meridional velocities,” in *LEFE/GMMC Scientific days*, online presentation, 2023.
- **D. Cortés-Morales** and A. Lazar, “Observation-based study of vertical velocities within the global thermocline: interannual variability and subsequent tracer fluxes,” in *TAV-PIRATA/TRIATLAS meeting*, online presentation, 2023.
- **D. Cortés-Morales** and A. Lazar, “Global open ocean vertical velocity reconstruction from observation-based geostrophic meridional velocity field,” in *AGU General Assembly*, in-situ poster, 2022.
- **D. Cortés-Morales** and A. Lazar, “North Atlantic thermocline vertical velocity reconstruction from observation-based geostrophic meridional velocity field,” in *EGU General Assembly*, in-situ presentation, 2022.
- **D. Cortés-Morales** and A. Lazar, “Estimates of large-scale vertical velocities in the tropical Atlantic Ocean.,” in *PIRATA-24/TAV*, online presentation, 2021.

Technical Reports

- I. Hernández-Carrasco, **D. Cortés-Morales**, V. Rossi, and C. Rwawi, “ECRs Technical Report,” 2025. [Online]. Available: http://ricerca.ismar.cnr.it/4DMED/Results.html.
- **D. Cortés-Morales** and I. Hernández-Carrasco, “4DMED Experimental Product Description (4DMED-EPD),” 2024. [Online]. Available: http://ricerca.ismar.cnr.it/4DMED/Results.html.

Evaluation of vorticity balance budget with CROCO simulation.

PEER REVIEW

Ocean Science

2025

Journal of Physical Oceanography

2024, 2025

TEACHING

B.Sc. Internship supervisor

IMEDEA

2025 Mallorca, Spain

Assessment of 30-year Mediterranean Sea eddy dataset from altimetry observations

B.Sc. Internship supervisor

IMEDEA

2024, 2025 Mallorca, Spain

Study of the Permanent and Variable Components of Kinetic Energy in the Global Ocean

Language Assistant

Sorbonne Université

2021 – 2023 Paris, France

English and Spanish conversation courses

OUTREACH

Outreach and Scientific Culture Program (PECI)

Mediterranean Institute of Advanced Studies, IMEDEA

Scientific advisor

2024 – 2026 Mallorca, Spain

Commission for communication and scientific dissemination

Mediterranean Institute of Advanced Studies, IMEDEA

Member

2024 – 2026 Mallorca, Spain

A Lagrangian View of Mediterranean Vortices over Three Decades of Satellite Altimetry

- I. Hernández-Carrasco, **D. Cortés-Morales**, N. Verbrugge, *et al.*, “Product Validation Report (PVR),” 2024. [Online]. Available: http://ricerca.ismar.cnr.it/4DMED/Results.html.

Databases produced

OLIV3

- **D. Cortés-Morales** and A. Lazar, *Global vertical velocity estimates from observation-based geostrophic meridional velocities over isopycnal levels*. DOI: https://zenodo.org/records/11506778.
- **D. Cortés-Morales** and A. Lazar, *Global vertical velocity estimates from observation-based geostrophic meridional velocities over vertical levels*. DOI: https://zenodo.org/records/11520783.
- **D. Cortés-Morales**, A. Lazar, D. Ruiz-Pino, and J. Mignot, *Global Observation-based LInear Vorticity Vertical Velocities (OLIV3) over vertical levels and monthly frequency*. DOI: https://doi.org/10.5281/zenodo.19469511.

Mediterranean FSLE

- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-size Lyapunov Exponents in the Mediterranean Sea derived from 4DVARNET8 geostrophic velocities (1/24°)*. DOI: https://zenodo.org/records/11612964.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-size Lyapunov Exponents in the Mediterranean Sea derived from 4DVARNET8 geostrophic velocities (1/72°)*. DOI: https://zenodo.org/records/11613002.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-size Lyapunov Exponents in the Mediterranean Sea derived from MIOST geostrophic velocities (1/24°)*. DOI: https://zenodo.org/records/11384073.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-size Lyapunov Exponents in the Mediterranean Sea derived from MIOST geostrophic velocities (1/72°)*. DOI: https://zenodo.org/records/11386592.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-size Lyapunov Exponents in the Mediterranean Sea derived from 4DVARNET20 geostrophic velocities (1/24°)*. DOI: https://zenodo.org/records/11610512.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-size Lyapunov Exponents in the Mediterranean Sea derived from 4DVARNET20 geostrophic velocities (1/72°)*. DOI: https://zenodo.org/records/11610889.

Mediterranean FTLV

- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-Time Lagrangian Vorticity in the Mediterranean Sea derived from 4DVARNET20 geostrophic velocities (1/24°)*. DOI: https://zenodo.org/records/13769355.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-Time Lagrangian Vorticity in the Mediterranean Sea derived from 4DVARNET20 geostrophic velocities (1/72°)*. DOI: https://zenodo.org/records/13771334.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-Time Lagrangian Vorticity in the Mediterranean Sea derived from 4DVARNET8 geostrophic velocities (1/24°)*. DOI: https://zenodo.org/records/11610512.

Mediterranean Institute of Advanced Studies, IMEDEA
Seminar
February 2026 — Mallorca, Spain

European Researchers’ Night
Universitat de les Illes Balears, UIB
Science fair participant
September 2025 — Mallorca, Spain

Ciència per a tothom
Universitat de les Illes Balears, UIB
Science fair organizer and participant
May 2024, 2025 — Mallorca, Spain

Conciencia GenZ
Mediterranean Institute of Advanced Studies, IMEDEA
Think tank scientific moderator
June 2024 — Mallorca, Spain

Fête de la Science
Sorbonne Université
Scientific fair student’s representative and participant
October 2022, 2023 — Paris, France

WOCEAN online workshop
Sorbonne Université
Workshop organizer and participant
September 2022 — Paris, France

Thematic month sessions
Sorbonne Université
Seminar series organizer
2021 – 2022 — Paris, France

REFERENCES

Dr. Ismael Hernández Carrasco
Researcher at IMEDEA (UIB-CSIC)
ihernandez@imedea.uib-csic.es

Dr. Ananda Pascual
Researcher at IMEDEA (UIB-CSIC)
ananda.pascual@imedea.uib-csic.es

- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-Time Lagrangian Vorticity in the Mediterranean Sea derived from 4DVARNET8 geostrophic velocities (1/72°)*. DOI: https://zenodo.org/records/13752898.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-Time Lagrangian Vorticity in the Mediterranean Sea derived from MIOST geostrophic velocities (1/24°)*. DOI: https://zenodo.org/records/13753841.
- **D. Cortés-Morales** and I. Hernández-Carrasco, *ESA 4DMED-Sea - Finite-Time Lagrangian Vorticity in the Mediterranean Sea derived from MIOST geostrophic velocities (1/72°)*. DOI: https://zenodo.org/records/13769110.